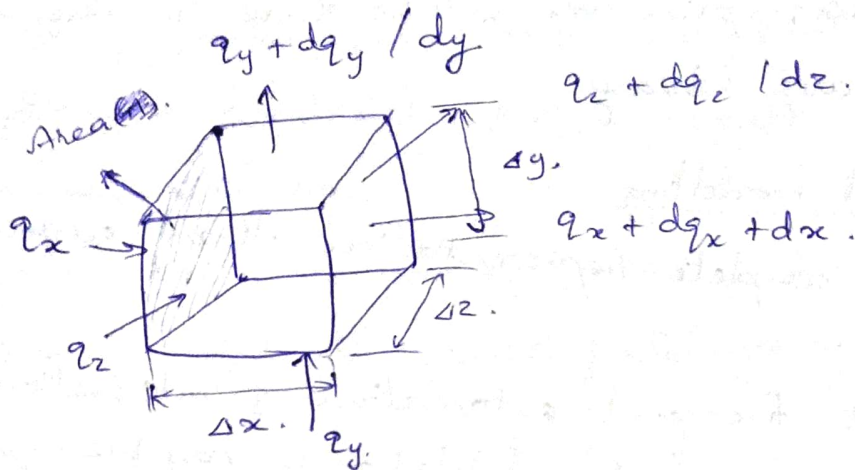


Elemental volume dx, dy, dz

Thermal conductivity k , density ρ , specific heat capacity c .



For solids $C_p \approx C_v$.

General heat conduction equation. ($\square$ coordinates).

$$T = T(x, y, z, \tau).$$

fourier law

Energy entering in x direc. $\Rightarrow Q_x = -kA \frac{\partial T}{\partial x}$

" leaving " $= Q_x + \Delta Q_x = Q_x + \frac{dQ_x}{dx} \Delta x.$

$$\Rightarrow -kA \frac{\partial T}{\partial x} + -kA \frac{\partial^2 T}{\partial x^2} \Delta x.$$

heat gen. per unit vol. $= \dot{q}$ (W/m^3).

Total heat gen $\Rightarrow Q_{gen} = \dot{q} \Delta x \Delta y \Delta z.$

Mass of elemental vol. $= \rho \Delta x \Delta y \Delta z.$

For energy leaving in y-dire. $\rightarrow -kA \frac{\partial T}{\partial y} + -kA \frac{\partial^2 T}{\partial y^2} \Delta y$

" z-direct $\rightarrow -kA \frac{\partial T}{\partial z} + -kA \frac{\partial^2 T}{\partial z^2} \Delta z$

Internal energy change of element, $m \Delta T = \rho C \Delta x \Delta y \Delta z \frac{\partial T}{\partial t}$

Energy balance $\Rightarrow$

$$Q_x + Q_y + Q_z + Q_{gen} = I \cdot E + Q_{x+\Delta x} + Q_{y+\Delta y} + Q_{z+\Delta z}$$

Internal heat generation $\Rightarrow$

when current is passing through the material,
nuclear fission etc.

General heat conduction eqn $\Rightarrow$

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} + \frac{\dot{q}}{k} = \frac{1}{\alpha} \frac{\partial T}{\partial t} = \frac{\rho C}{k} \frac{\partial T}{\partial t}$$

$$\alpha = \frac{k}{\rho C} \Rightarrow \text{Thermal diffusivity.}$$

when there's no heat generation, $\frac{\dot{q}}{k} = 0$.

then;

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} = \frac{1}{\alpha} \frac{\partial T}{\partial t}$$

$$\text{Steady state conduction; } \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} = 0$$

Assume ρ & C constant.

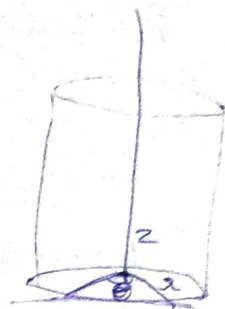
conductivity.

chance for
derivation.
Assignment
derivation.

Cylindrical / polar coordinates.

$$T = T(r, \theta, z)$$

$$\frac{\partial^2 T}{\partial r^2} + \frac{1}{r} \frac{\partial T}{\partial r} + \frac{1}{r^2} \frac{\partial^2 T}{\partial \theta^2} + \frac{\partial^2 T}{\partial z^2} = 0$$

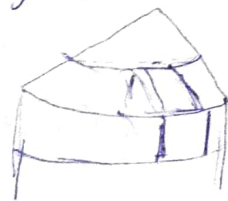


1D steady state

$$\frac{d^2 T}{dr^2} + \frac{1}{r} \frac{dT}{dr} = 0$$

heat transfer in radial direction only $\Rightarrow$

$$Q = -2\pi r k \Delta L \frac{dT}{dr}$$



HMT

James sir.

Heat transfer from fluid to solid wall or vice versa is called convection heat transfer.

$$Q = h_A A (T_A - T_s) \quad \text{--- (1)}$$

For conduction, $Q = kA \frac{(T_1 - T_2)}{\Delta x}$

$$Q = h_B A (T_2 - T_B) \quad \text{--- (2)}$$

(1) + (2) $\Rightarrow$

$$Q = \frac{T_A - T_B}{\frac{1}{h_A A} + \frac{\Delta x}{kA} + \frac{1}{h_B A}} \quad \Rightarrow \quad Q = UA \Delta T_{\text{overall}}$$

Overall heat transfer coefficient, $U = \frac{1}{\frac{1}{h_a} + \frac{\Delta x}{k} + \frac{1}{h_b}}$

$$Q = h_i A_i (T_i - T_1) \quad , \quad Q = h_o A_o (T_2 - T_o)$$

$$Q = \frac{2\pi k L (T_i - T_o)}{\ln(r_o/r_i)}$$

multiplying with $\frac{A_o}{A_o}$

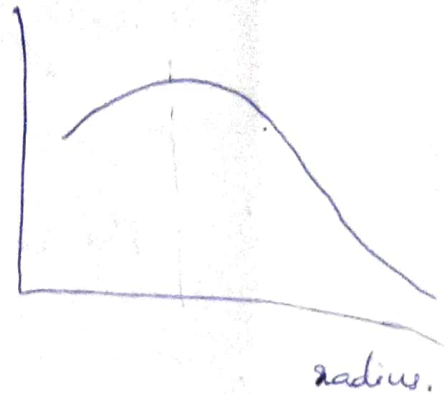
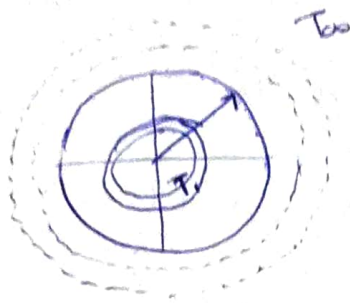
$$(a) \quad Q = \frac{T_i - T_o}{\frac{1}{h_i A_i} + \frac{\ln(r_o/r_i)}{2\pi k L} + \frac{1}{h_o A_o}} = \left[\frac{A_o (T_i - T_o)}{\frac{A_o}{A_i h_i} + \frac{A_o \ln(r_o/r_i)}{2\pi k L} + \frac{1}{h_o}} \right]$$

$$U_o = \frac{1}{\frac{A_o}{A_i} \frac{1}{h_i} + \frac{A_o \ln(r_o/r_i)}{2\pi k L} + \frac{1}{h_o}}$$

Multiplying (a) with $\frac{A_i}{A_i}$:

$$Q = \frac{A_i (T_i - T_o)}{\frac{1}{h_i} + \frac{A_i \ln(r_o/r_i)}{2\pi k L} + \frac{A_i}{A_o h_o}}$$

$$U_1 = \frac{1}{\frac{1}{h_i} + \frac{A_i \ln(r_o/r_i)}{2\pi k L} + \frac{A_i}{h_o r_o}}$$



$$Q = \frac{2\pi L (T_i - T_{\infty})}{\frac{\ln(r_o/r_i)}{k} + \frac{1}{h_o r_o}} \quad (f_{\infty})$$

$$\frac{dQ}{dr_o} = 0 \Rightarrow \frac{2\pi L (T_i - T_{\infty})}{\frac{\ln(r_o/r_i)}{k} + \frac{1}{r_o h}} = 0$$

$$\frac{-2\pi L (T_i - T_{\infty}) \left(\frac{1}{k r_o} - \frac{1}{h r_o^2} \right)}{\left(\frac{\ln(r_o/r_i)}{k} + \frac{1}{r_o h} \right)^2} = 0$$

$r_c = \frac{k}{h} \rightarrow$ critical radius of insulation.

Assignment (March-3) - 12:30 PM.

1. Write briefly about different types of heat exchangers with sketches.
2. Take the picture of an automobile radiator, explain its parts, identify the heat transfer elements, how the temperature of coolant is controlled.

3. Take the picture of a split air conditioner condenser, identify the heat transfer elements.
4. Take the picture of a solar collector & explain its working & heat transfer mechanisms.
5. Take the picture of a cooling tower & explain its working.

-
1. 1 face of a copper plate 10cm x 25cm x 3cm thick is maintained at 400°C. The other face is maintained at 100°C. Calculate the heat transfer rate through the plate (Ans: 925 kW).

Ans. $T_1 = 400^\circ\text{C} = 400 + 273 = \underline{673\text{K}}$.

$T_2 = 100^\circ\text{C} = 100 + 273 = \underline{373\text{K}}$.

Area = lbh = ~~10 x 25 x 3~~ $0.1 \times 0.25 \times 0.03 = \underline{0.00075\text{ m}^2}$.

$Q = kA \frac{dT}{dx}$.

$dT = 673 - 373 = \underline{300\text{K}}$.

$k = 386\text{ W/mK}$.

$\therefore Q = 386 \times 0.00075 \times \frac{300}{0.03} = \underline{96500} = \underline{96.5\text{ kW}}$.

2. An exterior of a house may be approx. by a 10.2cm layer of common brick ($k = 0.7\text{ W/m}^\circ\text{C}$) followed by a 3.8cm layer of gypsum ($k = 0.48\text{ W/m}^\circ\text{C}$). what thickness of

loosely packed rock wool insulation ($k = 0.065 \text{ W/m}^\circ\text{C}$) should be added to reduce the heat loss (or gain) through the wall by 80%. ? (Q with insulation / Q without insulation = 0.2) (Ans: 0.0584 m)

Ans. thickness = 10.2 cm = 0.102 m.

$$k_{\text{brick}} = 0.7 \text{ W/m}^\circ\text{C}$$

$$t_{\text{gypsum}} = 3.8 \text{ cm} = \underline{\underline{0.038 \text{ m}}}$$

$$k_{\text{gypsum}} = 0.48 \text{ W/m}^\circ\text{C}$$

$$k_{\text{rw}} = 0.065 \text{ W/m}^\circ\text{C}$$

$$\frac{Q_i}{Q} = 0.2$$